

PRD — MASJAVAS AI Cinematic Production & Post-Production Operating System

Version

v2.0 — Narration-First + Scene-Count-First + Reference Lock + Timeline Render Engine

Intake Summary

Pertanyaan	Jawaban User	Implikasi Produk
1. User sekarang menyelesaikan masalahnya dengan cara apa?	Banyak aplikasi terpisah	Produk harus menyatukan workflow ide, referensi, narasi, prompt, video, audio, subtitle, timeline, dan render.
2. Saat pertama kali buka aplikasi, user harus berhasil melakukan apa?	Buat proyek / konten / data pertama	Activation moment adalah user berhasil membuat project pertama dan menerima draft produksi awal.
3. Fitur paling wajib untuk MVP?	Generate AI, AI Assistant/ Chatbot, Upload file/gambar/ dokumen untuk referensi tokoh/karakter manual	MVP harus memiliki project workspace, upload reference, AI assistant, generation pipeline, dan reference lock.
4. Apa keunggulan dibanding cara lama?	Hasil lebih profesional	Sistem harus menghasilkan narasi, visual, audio, caption, timeline, dan render yang lebih cinematic dan konsisten.
5. Kenapa user akan balik lagi?	Bisa dipakai rutin dan menghemat waktu	Retention loop wajib berbasis project history, reusable references, saved style, template, render history, dan learning loop.

Product Doctrine

NARASI ADALAH NYAWA PRODUK.

MASJAVAS tidak boleh membuat storyboard, image prompt, video prompt, TTS, caption, timeline, atau final render sebelum Narration Master lolos validasi.

User memilih total scene.

Setiap scene default = 10 detik.

Total durasi = totalScene × 10 detik.

Ide dikonversi ke jumlah scene yang diminta user.
Semua video, audio, caption, dan timeline mengikuti scene plan tersebut.

Final Product Contract

MASJAVAS = AI Cinematic Production + Post-Production Operating System
GrokPI = Primary AI Provider Gateway

USER FIRST SUCCESS = Create first project/content
CORE INPUTS = Topic + Total Scenes + Tone + Platform + Mode + References
CORE MVP = Upload Reference + AI Assistant + Narration Master + Generate AI Pipeline + Timeline Render + Export Package
VALUE = Professional cinematic output
RETENTION = Routine use + time saving + reusable references + project history + render history

NO TOTAL SCENE → NO DURATION
NO STORYCORE → NO NARRATIVE ARC
NO NARRATIVE ARC → NO NARRATION DNA
NO NARRATION DNA → NO SCENE NARRATION
NO NARRATION MASTER PASS → NO STORYBOARD
NO NARRATION MASTER PASS → NO IMAGE PROMPT
NO NARRATION MASTER PASS → NO VIDEO PROMPT
NO NARRATION TIMING PASS → NO TTS
NO REFERENCE VALIDATION → NO VIDEO PROMPT
NO SCENE VIDEO ASSET → NO TIMELINE
NO AUDIO SYNC PASS → NO RENDER
NO CAPTION SYNC PASS → NO FINAL EXPORT
NO RENDER VALIDATION PASS → NO DOWNLOAD FINAL
NO GROKPI MODEL SYNC → NO PROVIDER ROUTING
NO SECRET MANAGER → NO PRODUCTION DEPLOYMENT

1. Overview

1.1 Ringkasan Produk

MASJAVAS AI adalah **AI-native cinematic production and post-production operating system** yang membantu creator mengubah satu ide menjadi film/video final siap render. Sistem ini tidak berhenti di prompt atau scene package, tetapi menyelesaikan seluruh alur produksi:

Ide
→ StoryCore
→ Narrative Arc
→ Narration Master

- Reference Set
- Scene Plan
- Storyboard
- Image Prompt
- Video Prompt
- TTS / Audio
- Scene Video
- Caption
- Timeline Assembly
- Audio Mixing
- Render Final
- Export Package

Produk ini menggantikan workflow manual yang biasanya dilakukan dengan banyak aplikasi: riset, nulis narasi, membuat prompt, generate video, generate audio, masuk CapCut, susun scene, sync narasi, tambah SFX, tambah subtitle, dan render final.

GrokPI menjadi **primary AI provider gateway** untuk model orchestration, text generation, image generation, image editing, video generation, image-to-video, TTS, quota, model routing, token pool, async job, usage logs, dan provider operations.

1.2 Masalah Utama

User saat ini memakai banyak aplikasi terpisah untuk membuat video AI. Masalah utamanya:

1. Ide, narasi, prompt, referensi, video, audio, dan subtitle tersebar di banyak tool.
2. Narasi sering terasa seperti AI: repetitif, generik, terlalu menjelaskan visual, moralizing, atau tidak cinematic.
3. Karakter, wajah, kostum, lokasi, dan visual style sering berubah antar scene.
4. Prompt image dan prompt video tercampur.
5. Durasi scene tidak konsisten.
6. Audio narasi, audio video asli, SFX, dan musik harus disusun manual.
7. Subtitle/caption harus dibuat dan disinkronkan manual.
8. User masih harus memakai CapCut untuk menggabungkan scene menjadi satu render final.
9. Tidak ada validation gate sebelum output buruk masuk tahap render.
10. Tidak ada observability atas provider, asset, timeline, audio sync, caption sync, dan render failure.

1.3 Target User

Segment	Problem	Primary Need
AI video creator	Banyak tool, output tidak konsisten	End-to-end workflow dari ide ke final render
YouTube documentary creator	Narasi harus kuat dan cinematic	Narration-first pipeline, source discipline, scene timing
Short-form creator	Butuh produksi cepat dan hook kuat	Scene-count-first, 10s scene, fast render

Segment	Problem	Primary Need
AI video producer	Karakter/lokasi drift	Universal Reference Set + Reference Lock
Creative agency kecil	Harus kirim output profesional ke klien	Project workspace, approval, render package
Solo creator	Tidak punya editor/post-production team	AI assistant + auto timeline + auto render

1.4 Business Objective

Membangun SaaS/workflow engine yang:

- Mengurangi ketergantungan pada banyak aplikasi terpisah.
- Menjadikan narasi sebagai kualitas utama produk.
- Menghasilkan video AI yang lebih profesional, konsisten, dan siap render.
- Mengotomatiskan kerja editing dasar yang biasanya dilakukan di CapCut.
- Menyediakan reusable reference, style, narration DNA, dan project template untuk pemakaian rutin.
- Membuka monetisasi melalui subscription, generation quota, render quota, storage, dan premium post-production tools.

1.5 Success Definition

Produk dianggap berhasil jika user dapat membuat project pertama, mengunci narasi, menghasilkan scene, menyusun timeline otomatis, dan mendapatkan final render tanpa harus menggabungkan manual di aplikasi lain.

Metric	MVP Target
Project activation rate	$\geq 75\%$
User mencapai Narration Master Lock	$\geq 70\%$
Time to first draft package	≤ 10 menit
Time to first draft render	≤ 30 menit untuk project pendek
Human-like narration score	≥ 95
Narration repetition score	≥ 95
Narration speakability score	≥ 95
Scene count accuracy	100%
Scene duration accuracy	100% scene = 10 detik
Reference consistency pass rate	$\geq 95\%$
Timeline assembly success	$\geq 95\%$
Audio sync pass rate	$\geq 95\%$

Metric	MVP Target
Caption sync pass rate	$\geq 95\%$
Render success rate	$\geq 95\%$
User-rated professional quality	$\geq 4/5$
Weekly repeat usage	$\geq 35\%$ beta users
Safety violation rate	$\leq 0.5\%$
Secret leakage incident	0

1.6 MVP Scope

MVP wajib mencakup:

1. Authentication + workspace.
2. Project creation berbasis TOPIC, TOTAL_SCENES, TONE, PLATFORM, MODE.
3. Upload file/gambar/dokumen untuk reference tokoh, karakter, style, lokasi, dan source.
4. AI Assistant / Project Copilot.
5. StoryCore Extractor.
6. Scene Count Adapter.
7. Narrative Arc Planner.
8. Narration DNA Builder.
9. Narration Master Engine.
10. Narration Quality Validator.
11. Universal Reference Set Generator.
12. Reference Prompt Generator.
13. Reference Validator.
14. Scene Plan + Beat Map Generator.
15. Storyboard Image Prompt Engine.
16. Image Prompt Engine.
17. Video Prompt Engine.
18. TTS Prompt / Audio Direction Engine.
19. Scene Master Engine.
20. GrokPI Provider Adapter.
21. Scene Video Asset Manager.
22. Auto Caption Engine.
23. Audio Mixing Engine.
24. Timeline Assembly Engine.
25. Render Composition Engine.
26. Render Validator.
27. Final MP4 Export.
28. CapCut Handoff Package.
29. Observability, audit logs, fallback, repair mode.

1.7 Out-of-Scope Awal

Tidak masuk MVP awal:

- Marketplace template publik.
- Direct publishing ke YouTube/TikTok/Instagram.
- Full collaborative editing seperti CapCut desktop.
- Advanced motion graphics editor.
- Multi-user enterprise approval kompleks.
- Advanced social analytics.
- Full legal/factual claim automation tanpa human review.
- Public multi-provider switching UI.
- Fully autonomous publishing tanpa user approval.

2. Requirements

2.1 Requirement Table

ID	Requirement	Priority	Owner	Acceptance Criteria
REQ-001	User dapat login/register dan membuat workspace.	Must	Engineering	User dapat masuk, session aman, workspace dibuat.
REQ-002	User dapat membuat project dari TOPIC, TOTAL_SCENES, TONE, PLATFORM, MODE.	Must	Product	Project tersimpan; total duration dihitung otomatis = totalScenes × 10.
REQ-003	Sistem menggunakan total scene sebagai input utama, bukan durasi bebas.	Must	Product	Jika user memilih 30 scene, sistem menghasilkan 30 scene × 10 detik.
REQ-004	Setiap scene default berdurasi 10 detik.	Must	System	SceneDurationValidator pass 100%.
REQ-005	User dapat upload file/gambar/ dokumen sebagai referensi karakter, tokoh, lokasi, style, source.	Must	Product/ Engineering	File tervalidasi, tersimpan, dan dapat dipakai Reference Set.
REQ-006	AI Assistant dapat membantu user menyusun brief, memperbaiki narasi, menjalankan repair, dan menjelaskan validation.	Must	AI/Product	Assistant project-aware dan tidak bisa bypass gate.

ID	Requirement	Priority	Owner	Acceptance Criteria
REQ-007	StoryCore Extractor mengekstrak characters, opposing forces, world, object affordances, story moments, dan scene intent.	Must	AI	StoryCore completeness ≥ 95 sebelum Narrative Arc.
REQ-008	Narrative Arc Planner membagi cerita sesuai total scene.	Must	AI	Setiap scene punya narrative function.
REQ-009	Narration DNA Builder membuat style suara narator sebelum narasi scene.	Must	AI	Narration DNA approved/ locked sebelum scene narration.
REQ-010	Narration Master Engine membuat narasi per scene yang human-like, cinematic, tidak repetitif, dan timing-safe.	Must	AI/ Narration	Human-like ≥ 95 , repetition ≥ 95 , speakability ≥ 95 .
REQ-011	Narasi wajib lolos sebelum storyboard/prompt/video/ audio/caption/render dibuat.	Must	Workflow	Downstream phase diblokir jika Narration Master belum pass.
REQ-012	Narasi tidak boleh hanya mendeskripsikan visual.	Must	Narration	Visual-description score berada di bawah threshold.
REQ-013	Narasi harus mulai tepat detik 2.0 per scene.	Must	Audio/ Timing	AudioStartValidator pass 100%.
REQ-014	Dialog hanya boleh setelah 5.5 detik.	Must	Audio/ Timing	DialogueTimingValidator pass 100%.
REQ-015	Universal Reference Set dibuat otomatis dari StoryCore, bukan hardcoded template.	Must	AI/ Reference	Reference set 6–8 item dibuat dari StoryCore.
REQ-016	Reference set mendukung PrimaryCharacterRef, AlternateStateCharacterRef, SupportingCharacterRef, OpposingForceRef, GroupOrInstitutionRef, PropRef, CreatureRef, EnvironmentRef.	Must	AI/ Reference	Role coverage pass.
REQ-017	Character reference hanya untuk identity lock.	Must	Media	ReferenceRoleValidator pass.
REQ-018	Storyboard reference hanya untuk struktur visual dan flow.	Must	Media	Tidak ada storyboard grid/panel/text di video prompt.

ID	Requirement	Priority	Owner	Acceptance Criteria
REQ-019	Environment reference hanya untuk lokasi, lighting, material, mood.	Must	Media	Environment role compliance pass.
REQ-020	Previous video reference hanya untuk continuity.	Must	Media	Previous video tidak membuat story restart.
REQ-021	Scene Plan wajib memiliki beat map 0–10 detik.	Must	Scene Engine	Semua scene memiliki 5 beat standar.
REQ-022	Beat map standar: 0–2 visual hook, 2–5.5 narration, 5.5–7 dialogue/reaction, 7–8.8 emotional detail, 8.8–10 transition.	Must	Scene Engine	BeatMapValidator pass.
REQ-023	Image Prompt dan Video Prompt wajib dipisahkan.	Must	Media	Image/video separation = 100%.
REQ-024	Video Prompt membuat 10-second event, bukan deskripsi gambar statis.	Must	Media	Motion and timing metadata complete.
REQ-025	TTS Prompt mengikuti Narration DNA, timing, pause, emotion, dan pacing.	Must	Audio	TTS readiness ≥ 95 .
REQ-026	GrokPI menjadi provider utama untuk text, image, image edit, video, image-to-video, TTS.	Must	Platform	Adapter mendukung endpoint utama.
REQ-027	Async video generation dikelola dengan queued, processing, completed, failed, timeout.	Must	Platform	Video job state handling pass.
REQ-028	Scene Video Asset Manager menyimpan dan memvalidasi video per scene.	Must	Media	Semua scene punya asset atau render diblokir.
REQ-029	Timeline Assembly Engine menggabungkan scene video secaraurut berdasarkan sceneNumber.	Must	Post-Production	Timeline urut, tidak ada scene hilang.
REQ-030	Audio Mixing Engine menggabungkan narration, original scene audio, dialogue, SFX, dan music.	Must	Post-Production	Audio sync pass; clipping = 0.
REQ-031	Auto Caption Engine membuat SRT/VTT/ASS/JSON dari narration dan dialogue.	Must	Post-Production	Caption sync pass ≥ 95 .

ID	Requirement	Priority	Owner	Acceptance Criteria
REQ-032	Subtitle tidak boleh muncul di raw generated video; subtitle hanya ditambahkan saat final render jika user memilih.	Must	Media/Post	Raw video prompt no visible text; final caption optional.
REQ-033	Render Composition Engine menghasilkan final MP4.	Must	Post-Production	Final duration = totalScenes × 10.
REQ-034	CapCut Handoff Export membuat folder asset terstruktur.	Should	Export	Folder berisi scene videos, audio, subtitles, timeline JSON, manifest.
REQ-035	Export hanya boleh dibuat jika validation pass.	Must	Release	No validation pass = no export.
REQ-036	Semua provider secret disimpan server-side.	Must	Security	Tidak ada API key di frontend, logs, export, PDF.
REQ-037	Semua critical events dicatat.	Must	Reliability	Critical path telemetry coverage = 100%.
REQ-038	Project history dan reusable references tersedia untuk repeat usage.	Should	Product	User bisa reuse character/style/reference di project baru.
REQ-039	Feedback loop menyimpan user rating dan repair signal.	Should	Product/AI	Feedback masuk learning profile.
REQ-040	Prompt/model/provider/render config memiliki rollback switch.	Must	Release	Rollback bisa dilakukan jika threshold gagal.

3. Core Features

3.1 Project Workspace & Scene-Count-First Project Builder

Description:

Workspace untuk membuat project video berdasarkan jumlah scene, bukan durasi bebas.

User Story:

Sebagai creator, saya ingin memilih jumlah scene agar aplikasi mengubah ide saya menjadi struktur cerita yang sesuai durasi pasti.

Input:

- Topic
- Total scenes
- Tone
- Platform

- Mode
- References optional

Output:

- Project record
- Total duration
- Pipeline initialized

Business Rules:

- Total duration = totalScenes × 10 detik.
- totalScenes minimum dan maksimum mengikuti plan.
- durationPerSceneSec default = 10.

Edge Cases:

- User meminta durasi bebas.
- User memilih scene terlalu banyak untuk ide pendek.
- User memilih scene terlalu sedikit untuk cerita kompleks.

Error States:

- Invalid totalScenes
- Unsupported platform
- Missing topic

Acceptance Criteria:

- [] User dapat membuat project pertama.
- [] Total duration dihitung otomatis.
- [] Pipeline menampilkan jumlah scene dan target durasi final.

3.2 Upload Reference File / Image / Document

Description:

Upload referensi tokoh, karakter, wajah, kostum, lokasi, style, dokumen source, dan moodboard.

User Story:

Sebagai AI video producer, saya ingin upload reference agar karakter dan visual tidak berubah antar scene.

Input:

- Image reference
- Document/source file
- Character note
- Location/style reference

Output:

- Stored reference asset
- Reference metadata
- Candidate role assignment

Business Rules:

- File harus divalidasi tipe, ukuran, permission, dan safety.
- Reference harus dikaitkan ke project.
- Sensitive content harus dimasking bila masuk model call.

Edge Cases:

- Gambar buram.
- Dokumen tidak terbaca.
- Reference bertentangan dengan StoryCore/World Bible.

Error States:

- Unsupported file
- Unsafe file
- Low-quality reference
- Permission denied

Acceptance Criteria:

- [] File berhasil diupload dan diberi label.
 - [] File bisa dipakai dalam Universal Reference Set.
 - [] Reference gagal memberi pesan actionable.
-

3.3 AI Assistant / Project Copilot

Description:

AI assistant yang membantu user menyusun brief, memperbaiki narasi, menjelaskan validation, dan menjalankan repair.

User Story:

Sebagai creator, saya ingin dibimbing oleh AI agar saya tidak harus memahami semua pipeline teknis.

Input:

- User question
- Project state
- Current phase
- Validation report

Output:

- Answer

- Repair suggestion
- Next action
- Clarification question

Business Rules:

- Assistant tidak boleh bypass gate.
- Assistant harus project-aware dan permission-aware.
- Assistant harus menjelaskan low-confidence state dengan bahasa mudah.

Edge Cases:

- User meminta export walau gagal.
- User meminta bypass narasi.
- User memasukkan instruksi ambigu.

Error States:

- Low confidence
- Unsafe request
- Permission denied

Acceptance Criteria:

- [] Assistant dapat menjelaskan status pipeline.
- [] Assistant dapat memulai Narration Repair Mode.
- [] Assistant tidak dapat melewati validation gate.

3.4 StoryCore Extractor

Description:

Mengekstrak inti cerita dari topic dan reference user.

User Story:

Sebagai creator, saya ingin ide mentah saya dipahami menjadi struktur cerita yang kuat.

Input:

- Topic
- Uploaded references
- Mode
- Platform

Output:

- AIExtractedStoryCore

StoryCore Fields:

- characters
- protagonist

- opposing forces
- world
- emotional core
- conflict
- transformation
- object affordances
- story moments
- possible scene functions
- risk flags

Business Rules:

- StoryCore harus dibuat sebelum Narrative Arc.
- StoryCore tidak boleh hardcode karakter tertentu.
- StoryCore harus universal untuk semua topik.

Acceptance Criteria:

- [] StoryCore completeness ≥ 95 .
- [] Semua entity penting diberi ID.
- [] Risk flags tercatat.

3.5 Narrative Arc Planner

Description:

Mengubah StoryCore menjadi struktur cerita sesuai total scene.

User Story:

Sebagai creator, saya ingin ide saya dipecah menjadi alur scene yang proporsional.

Input:

- StoryCore
- totalScenes
- tone
- platform

Output:

- Narrative Arc Map
- Scene function per scene

Business Rules:

- Setiap scene harus punya fungsi naratif.
- Scene tidak boleh repetitif secara fungsi.
- Arc harus punya hook, escalation, turning point, reveal/payoff, closing.

Acceptance Criteria:

- [] Semua scene punya sceneFunction.
 - [] Hook dan payoff terhubung.
 - [] Arc pass sebelum Narration DNA.
-

3.6 Narration DNA Builder

Description:

Membuat identitas suara narasi sebelum scene narration dibuat.

User Story:

Sebagai creator, saya ingin narasi memiliki gaya suara yang konsisten, profesional, dan tidak terasa AI.

Input:

- StoryCore
- Narrative Arc
- Tone
- Platform
- Audience

Output:

- Narration DNA

Narration DNA Includes:

- narrator persona
- emotional distance
- moral explicitness
- metaphor level
- sentence length preference
- forbidden words
- preferred vocabulary
- dialogue frequency
- pacing rules
- TTS direction

Business Rules:

- Narration DNA wajib locked sebelum scene narration.
- Narration DNA harus menjaga genre consistency.

Acceptance Criteria:

- [] Narration DNA dibuat dan tersimpan.
 - [] Avoid words dan preferred words tersedia.
 - [] TTS direction tersedia.
-

3.7 Narration Master Engine

Description:

Engine utama untuk membuat, memperbaiki, menilai, dan mengunci narasi profesional sebelum artefak lain dibuat.

User Story:

Sebagai creator, saya ingin narasi yang terasa ditulis manusia profesional, cinematic, tidak repetitif, tidak melanggar aturan, dan siap dibacakan.

Input:

- StoryCore
- Narrative Arc
- Narration DNA
- totalScenes
- platform
- tone

Output:

- Scene narration list
- Dialogue policy
- Voice direction
- TTS pacing map
- Narration validation report
- Narration Master Lock

Business Rules:

- Narasi dibuat sebelum storyboard, image prompt, video prompt, TTS, caption, timeline, dan render.
- Narasi bukan visual description.
- Narasi harus membawa emosi, subtext, tension, atau meaning.
- Narasi harus human-like.
- Narasi tidak boleh repetitif.
- Narasi harus speakable.
- Narasi harus timing-safe.
- Narasi harus policy-safe.

Narration Law:

1. Narasi bukan deskripsi visual.
2. Narasi harus membawa emosi, makna, atau ketegangan.
3. Satu scene = satu gagasan emosional.
4. Jangan ulang struktur kalimat yang sama lebih dari 2 kali.
5. Jangan ulang kata kunci emosional berlebihan.
6. Jangan gunakan bahasa generik AI.
7. Jangan overdramatic.
8. Jangan menjelaskan yang sudah jelas terlihat di visual.

9. Jangan membuat klaim faktual tanpa source.
10. Jangan membuat dialog jika scene cukup kuat dengan silence.
11. Narasi harus speakable.
12. Narasi harus punya jeda alami.
13. Narasi harus cocok untuk TTS.
14. Narasi harus menjaga rasa penasaran antar scene.
15. Narasi final harus lolos quality score sebelum visual dibuat.

Quality Metrics:

Metric	Target
Human-like quality	≥ 95
Emotional clarity	≥ 95
Speakability	≥ 95
Non-repetitive score	≥ 95
Scene specificity	≥ 95
Subtext quality	≥ 90
Genre consistency	100
Policy/safety pass	100
Timing fit	≥ 95
TTS readiness	≥ 95

Error States:

- Sounds like AI
- Too repetitive
- Too visual-descriptive
- Too generic
- Too moralizing
- Too long for timing
- Policy violation
- Genre leak

Acceptance Criteria:

- [] Narration Master Lock hanya dibuat jika semua metric pass.
- [] Jika gagal, masuk Narration Repair Mode.
- [] Downstream artifact generation diblokir sebelum lock.

3.8 Anti-AI Tone Detector

Description:

Validator untuk mendeteksi gaya tulisan yang terasa seperti AI.

Detects:

- generic phrasing
- repeated structure
- over-explaining
- moral lecture
- unnatural Indonesian
- overdramatic wording
- visual description overload
- template-like sentence
- genre mismatch

Acceptance Criteria:

- [] Narasi dengan AI-like score tinggi ditolak.
 - [] Sistem memberi alasan dan contoh perbaikan.
 - [] Repair loop mengurangi AI-like score.
-

3.9 Repetition Detector

Description:

Mendeteksi repetisi kata, struktur, emosi, dan fungsi scene.

Checks:

- repeated opening phrase
- repeated emotional word
- repeated sentence pattern
- repeated scene function
- repeated moral statement
- repeated hook device

Acceptance Criteria:

- [] Repetition score ≥ 95 .
 - [] Repetisi lintas scene ditandai.
 - [] Repair suggestion tersedia.
-

3.10 Narration Timing Validator

Description:

Memastikan narasi cocok dengan scene 10 detik dan mulai detik 2.0.

Timing Rule:

0.0–2.0s = visual hook only
2.0–5.5s = main narration phrase

5.5–7.0s = optional dialogue/reaction
7.0–8.8s = emotional echo / silence / short continuation
8.8–10.0s = transition

Acceptance Criteria:

- [] Narasi tidak mulai sebelum 2.0s.
- [] Dialog tidak mulai sebelum 5.5s.
- [] Estimated read time cocok dengan timing budget.
- [] TTS-ready pacing tersedia.

3.11 Universal Reference Set Generator

Description:

Membuat 6–8 reference utama otomatis dari StoryCore, bukan dari template hardcoded.

User Story:

Sebagai creator, saya ingin reference tokoh/karakter/objek/lokasi dibuat otomatis sesuai cerita saya.

Input:

- AIExtractedStoryCore
- Uploaded references
- Narrative Arc
- Scene Plan

Output:

- Universal Reference Set
- Reference prompts
- Validation report

Reference Roles:

- PrimaryCharacterRef
- AlternateStateCharacterRef
- SupportingCharacterRef
- OpposingForceRef
- GroupOrInstitutionRef
- PropRef
- CreatureRef
- EnvironmentRef

Business Rules:

- Planner memilih maksimal 6–8 reference utama.
- Reference harus berasal dari StoryCore.
- Character reference hanya identity lock.
- Environment reference hanya location/mood/material/lighting lock.

- Storyboard reference hanya struktur visual.
- Previous video reference hanya continuity.

Acceptance Criteria:

- [] Role coverage pass.
- [] Identity lock pass.
- [] Provider-ready pass.
- [] Tidak ada special-case hardcoded.

3.12 Reference Prompt Generator

Description:

Membuat prompt image reference untuk character, prop, environment, creature, atau institution/group.

Input:

- Universal Reference Item
- World context
- Visual DNA

Output:

- Image reference prompt
- Negative prompt
- Usage rule

Business Rules:

- Reference prompt tidak boleh menjadi storyboard/action scene.
- CharacterRef wajib full-body reference sheet atau identity sheet.
- EnvironmentRef wajib location/material/lighting/mood lock.
- PropRef wajib object identity and affordance lock.

Acceptance Criteria:

- [] Prompt sesuai role.
- [] Negative prompt tersedia.
- [] Usage rules tersedia.

3.13 Scene Plan & Beat Map Generator

Description:

Membuat scene plan berbasis totalScenes dan Narration Master.

Input:

- Narration Master

- Narrative Arc
- totalScenes
- Reference Set

Output:

- Scene plan
- Beat map per scene
- Story action per scene

Business Rules:

- 1 scene = 10 detik.
- Setiap scene punya 5 beat standar.
- Scene plan mengikuti narasi, bukan sebaliknya.

Acceptance Criteria:

- [] Semua scene punya beat map.
- [] Beat map mengikuti timing rule.
- [] Scene plan pass sebelum storyboard.

3.14 Storyboard Image Prompt Engine

Description:

Membuat prompt storyboard image untuk struktur visual 10 detik per scene.

Output Format:

- 1 hero frame utama
- 5 panel kecil berurutan

Business Rules:

- Storyboard dipakai sebagai struktur visual, bukan sebagai raw video look.
- Storyboard boleh punya label minimal untuk produksi, tetapi raw video prompt harus melarang visible text.
- Storyboard dibuat setelah Narration Master pass.

Acceptance Criteria:

- [] Storyboard prompt dibuat per scene.
- [] Timing 0–10 detik tercermin.
- [] Tidak menjadi final video prompt.

3.15 Image Prompt Engine

Description:

Membuat keyframe prompt per scene menggunakan Reference Set.

Business Rules:

- Image prompt = keyframe, bukan motion event.
- Wajib memakai reference ID.
- Tidak boleh mencampur storyboard labels ke image final.

Acceptance Criteria:

- [] Image prompt memakai reference ID.
 - [] Negative prompt tersedia.
 - [] Reference compliance pass.
-

3.16 Video Prompt Engine

Description:

Membuat prompt video sebagai 10-second event dengan motion, camera, timing, audio rule, reference role, dan continuity.

Video Prompt Must Include:

- duration 10 seconds
- audio timing rule
- references and role use
- story action
- beat map
- narration
- dialogue if any
- camera
- motion
- sound
- negative

Business Rules:

- No narration/dialogue before 2.0s.
- Dialogue only after 5.5s.
- No subtitles or visible text in raw video.
- Previous video only for continuity.
- Video prompt must not show storyboard grid, labels, panels, captions, arrows.

Acceptance Criteria:

- [] Video prompt per scene generated.
 - [] Reference role pass.
 - [] Timing pass.
 - [] Visible text pass.
-

3.17 TTS Prompt & Audio Direction Engine

Description:

Membuat arahan suara untuk narasi, dialog, emotion, pause, pacing, dan pronunciation.

Input:

- Narration Master
- Narration DNA
- Scene timing

Output:

- TTS prompt
- Voice direction
- Audio timing map

Acceptance Criteria:

- [] TTS starts at 2.0s per scene.
 - [] Voice style konsisten.
 - [] Audio validation score ≥ 95 .
-

3.18 Scene Master Engine

Description:

Mengabungkan narration, reference, prompt, audio direction, video instruction, dan validation menjadi production sheet per scene.

Scene Master Includes:

- sceneNumber
- title
- durationSec = 10
- references
- storyAction
- beatMap
- narration
- dialogue
- camera
- motion
- sound
- imagePrompt
- videoPrompt
- ttsPrompt
- validation

Acceptance Criteria:

- [] Semua scene punya Scene Master.
 - [] Scene validation score ≥ 97 .
 - [] Scene gagal masuk Repair Mode.
-

3.19 GrokPI Provider Adapter

Description:

Adapter OpenAI-compatible untuk model sync, text/orchestration, image, image edit, video, TTS, async jobs, error mapping, and telemetry.

Endpoints:

- GET /v1/models
- POST /v1/chat/completions
- POST /v1/video/generations
- GET /v1/video/generations/{jobId}
- GET /v1/video/generations/{jobId}/result
- POST /v1/audio/speech

Acceptance Criteria:

- [] Model sync pass.
 - [] Text generation pass.
 - [] Video async lifecycle handled.
 - [] TTS generation pass.
 - [] Secrets never logged.
-

3.20 Scene Video Asset Manager

Description:

Menyimpan, mengelola, dan memvalidasi video per scene.

Input:

- Generated scene video
- Scene metadata
- Provider result

Output:

- Scene video asset
- Asset manifest
- Validation status

Acceptance Criteria:

- [] Semua scene video diberi sceneNumber.
 - [] Durasi video divalidasi 10 detik.
 - [] Missing asset memblokir timeline.
-

3.21 Timeline Assembly Engine

Description:

Menggabungkan video scene secara otomatis berdasarkan sceneNumber.

Input:

- Scene video assets
- Scene metadata
- totalScenes
- durationPerSceneSec

Output:

- Timeline JSON
- Ordered video track
- Draft timeline

Business Rules:

- Scene harus urut dari 1 sampai totalScenes.
- Tidak boleh ada scene hilang.
- Final duration = totalScenes × 10.

Acceptance Criteria:

- [] Scene order pass.
 - [] Missing scene blocked.
 - [] Timeline JSON dibuat.
-

3.22 Audio Mixing Engine

Description:

Menggabungkan narration audio, original scene audio, dialogue, sound effects, dan background music.

Audio Tracks:

Track	Type	Rule
A1	Narration	Mulai detik 2.0 per scene.
A2	Original scene audio	Boleh mulai 0.0, duck saat narasi aktif.

Track	Type	Rule
A3	Sound effects	Mengikuti beat map.
A4	Background music	Continuous, auto duck saat narasi/dialog.
A5	Dialogue	Hanya setelah 5.5 detik.

Mixing Features:

- loudness normalization
- auto ducking
- fade in/out
- crossfade
- limiter
- silence detection
- clipping detection

Acceptance Criteria:

- [] Narration starts at 2.0s.
- [] Dialogue starts ≥ 5.5 s.
- [] Clipping = 0.
- [] Loudness validator pass.

3.23 Auto Caption & Subtitle Engine

Description:

Membuat subtitle dari narration dan dialogue.

Sources:

- Primary: Narration text per scene
- Secondary: Dialogue text per scene
- Fallback: Speech-to-text from final audio

Output:

- SRT
- VTT
- ASS
- Caption JSON
- Burn-in caption optional

Business Rules:

- Raw generated video must contain no subtitles.
- Subtitle hanya ditambahkan saat final render jika user memilih.
- Caption sync mengikuti narration/dialogue timing.

Acceptance Criteria:

- [] SRT/VTT dibuat.
 - [] Caption sync pass ≥ 95 .
 - [] Burn-in optional bekerja.
-

3.24 Render Composition Engine

Description:

Merender scene videos, mixed audio, SFX, music, dan caption menjadi satu final MP4.

Input:

- Timeline JSON
- Video track
- Audio mix
- Caption track
- Render settings

Output:

- final_render.mp4
- final_audio_mix.wav
- subtitles.srt/vtt/ass
- render_report.json

Business Rules:

- Render hanya jika timeline, audio, caption, and asset validation pass.
- Final duration must equal totalScenes $\times 10$.

Acceptance Criteria:

- [] Final MP4 dibuat.
 - [] Final duration pass.
 - [] Render report PASS.
-

3.25 CapCut Handoff Export

Description:

Membuat folder asset yang rapi untuk user yang tetap ingin edit manual di CapCut.

Folder Structure:

```
/capcut_export_package/  
├─ 01_video_scenes/  
│   └─ scene_001.mp4
```

```

|   |─ scene_002.mp4
|   |─ ...
|─ 02_audio/
|   |─ narration.wav
|   |─ music.wav
|   |─ sfx_scene_001.wav
|   |─ ...
|─ 03_subtitles/
|   |─ subtitles.srt
|   |─ subtitles.vtt
|   |─ captions.json
|─ 04_timeline/
|   |─ timeline.json
|   |─ edit_decision_list.csv
|   |─ asset_manifest.json
|─ 05_final/
|   |─ final_render.mp4

```

Acceptance Criteria:

- [] Folder export dibuat.
- [] Naming file konsisten.
- [] Timeline JSON dan asset manifest tersedia.

4. User Flow

4.1 Main Happy Path

1. User login.
2. User membuat project baru.
3. User mengisi TOPIC, TOTAL_SCENES, TONE, PLATFORM, MODE.
4. Sistem menghitung $\text{totalDuration} = \text{totalScenes} \times 10$.
5. User upload reference file/gambar/dokumen.
6. AI Assistant membantu memperjelas brief jika perlu.
7. StoryCore Extractor membuat AIExtractedStoryCore.
8. Narrative Arc Planner membagi cerita sesuai total scene.
9. Narration DNA Builder membuat suara narator.
10. Narration Master Engine membuat narasi per scene.
11. Anti-AI Tone Detector, Repetition Detector, Policy Guardrail, Speakability Validator, dan Timing Validator memeriksa narasi.
12. Jika gagal, Narration Repair Mode berjalan.
13. Jika pass, Narration Master Lock dibuat.
14. Universal Reference Set Generator membuat 6–8 reference utama.
15. Reference Prompt Generator membuat prompt image reference.
16. Reference Validator memvalidasi role dan identity lock.
17. Scene Plan + Beat Map Generator membuat scene 10 detik.
18. Storyboard Image Prompt Engine membuat storyboard per scene.
19. Image Prompt Engine membuat keyframe prompt.

20. Video Prompt Engine membuat 10-second video prompt.
21. TTS Prompt Engine membuat audio direction.
22. User review dan approve Scene Master.
23. GrokPI Provider Adapter menjalankan generation.
24. Scene Video Asset Manager menyimpan video per scene.
25. Timeline Assembly Engine mengurutkan video scene.
26. Audio Mixing Engine menyusun narasi, audio asli, SFX, music, dialogue.
27. Auto Caption Engine membuat SRT/VTT/ASS/JSON.
28. Render Composition Engine membuat final MP4.
29. Render Validator memvalidasi final duration, audio sync, caption sync, completeness.
30. Export Engine menyediakan final render dan CapCut Handoff Package.
31. Feedback dan render history tersimpan.

4.2 Narration Repair Path

1. Narration validator gagal.
2. Sistem menampilkan alasan:
 3. terlalu AI
 4. terlalu repetitif
 5. terlalu visual
 6. terlalu panjang
 7. tidak speakable
 8. genre leak
9. AI Assistant menawarkan opsi:
 10. rewrite all
 11. repair selected scene
 12. change narrator style
 13. reduce moralizing
 14. make more cinematic
 15. simplify for TTS
16. Sistem menjalankan Script Doctor.
17. Narasi divalidasi ulang.
18. Jika pass, Narration Master Lock dibuat.

4.3 Reference Repair Path

1. Reference Validator gagal.
2. Sistem menandai role yang bermasalah.
3. User dapat upload reference tambahan atau memilih text-only reference.
4. Reference Planner memperbaiki set.
5. Provider dry-run dilakukan jika perlu.

4.4 Render Repair Path

1. Render Validator gagal.

2. Sistem menunjukkan penyebab:
3. missing scene video
4. scene duration mismatch
5. narration starts before 2s
6. dialogue before 5.5s
7. caption out of sync
8. audio clipping
9. final duration mismatch
10. User memilih:
11. regenerate failed scene
12. trim/stretch scene
13. rebuild audio mix
14. rebuild captions
15. render draft again
16. Render ulang hanya bagian yang gagal jika memungkinkan.

4.5 Low-Confidence Path

1. Confidence score di bawah threshold.
2. UI menampilkan label "Needs Review" atau "Could Not Verify".
3. Sistem menjelaskan alasan.
4. User dapat menambah reference, memperbaiki narasi, atau request human review.
5. Export final diblokir sampai validation pass.

5. Architecture

5.1 System Architecture

```
flowchart TD
    A[User / Creator] --> B[Frontend Workspace]
    A --> CHAT[AI Assistant Chat UI]
    B --> API[Backend API]
    CHAT --> API
    API --> AUTH[Authentication]
    AUTH --> RBAC[Authorization / RBAC]
    RBAC --> PROJ[Project Service]
    PROJ --> UP[Upload Reference Service]
    UP --> STORE[(Object Storage)]
    UP --> META[Reference Metadata Extractor]
    PROJ --> VALID[Input Validation Layer]
```

```

META --> VALID
VALID --> ORCH[AI Orchestration State Machine]

ORCH --> SCENECOUNT[Scene Count Adapter]
SCENECOUNT --> STORYCORE[StoryCore Extractor]
STORYCORE --> ARC[Narrative Arc Planner]
ARC --> DNA[Narration DNA Builder]
DNA --> NARR[Narration Master Engine]

NARR --> AITONE[Anti-AI Tone Detector]
AITONE --> REP[Repetition Detector]
REP --> POLICY[Narration Policy + Genre Guardrails]
POLICY --> TIMING[Narration Timing + Speakability Validator]
TIMING --> NLOCK{Narration Master Pass?}
NLOCK -->|No| NREPAIR[Narration Repair Mode]
NREPAIR --> NARR
NLOCK -->|Yes| NMASTER[Narration Master Lock]

NMASTER --> REFPLAN[Universal Reference Set Generator]
REFPLAN --> REF PROMPT[Reference Prompt Generator]
REF PROMPT --> REFVAL[Reference Validator]
REFVAL --> REFLOCK{Reference Pass?}
REFLOCK -->|No| RREPAIR[Reference Repair Mode]
RREPAIR --> REFPLAN
REFLOCK -->|Yes| SCENEPLAN[Scene Plan + Beat Map Generator]

SCENEPLAN --> STORYBOARD[Storyboard Image Prompt Engine]
SCENEPLAN --> IMAGE[Image Prompt Engine]
SCENEPLAN --> VIDEO[Video Prompt Engine]
SCENEPLAN --> TTS[TTS Prompt Engine]

STORYBOARD --> SCENEM[Scene Master Engine]
IMAGE --> SCENEM
VIDEO --> SCENEM
TTS --> SCENEM
SCENEM --> APPROVAL[User Approval Gate]
APPROVAL -->|Revise| NREPAIR
APPROVAL -->|Approved| GP[GrokPI Provider Adapter]

GP --> MODELS[GET /v1/models]
GP --> CHATAPI[POST /v1/chat/completions]
GP --> VIDEOAPI[POST /v1/video/generations]
GP --> SPEECH[POST /v1/audio/speech]
VIDEOAPI --> ASYNC[Async Video Poller]

CHATAPI --> ASSET[Media Asset Manager]
ASYNC --> ASSET
SPEECH --> AUDIOASSET[Audio Asset Manager]

ASSET --> TIMELINE[Timeline Assembly Engine]
AUDIOASSET --> MIX[Audio Mixing Engine]

```

```

SCENEM --> CAPTION[Auto Caption Engine]

TIMELINE --> RENDER[Render Composition Engine]
MIX --> RENDER
CAPTION --> RENDER

RENDER --> RVAL[Render Validator]
RVAL --> RPASS{Render Pass?}
RPASS -->|No| RENDERREPAIR[Render Repair Mode]
RENDERREPAIR --> RENDER
RPASS -->|Yes| FINAL[Final MP4 Export]

FINAL --> HANDOFF[CapCut Handoff Package]
FINAL --> DL[Download / Share / Save]

PROJ --> DB[(PostgreSQL Domain DB)]
ORCH --> DB
NARR --> DB
REFPLAN --> DB
SCENEM --> DB
GP --> DB
RENDER --> DB

API --> OBS[Telemetry / Logs / Traces]
ORCH --> OBS
NARR --> OBS
GP --> OBS
RENDER --> OBS
RVAL --> OBS

```

5.2 Narration-First Architecture

flowchart TD

```

A[User Topic + Total Scenes + Tone + Platform] --> B[StoryCore Extractor]
B --> C[Narrative Arc Planner]
C --> D[Narration DNA Builder]
D --> E[Scene Narration Draft Generator]
E --> F[Script Doctor Rewrite]
F --> G[Anti-AI Tone Detector]
G --> H[Repetition Detector]
H --> I[Policy + Genre Guardrails]
I --> J[Speakability + Timing Validator]
J --> K{Narration Master Pass?}
K -->|No| L[Narration Repair Loop]
L --> E
K -->|Yes| M[Narration Master Lock]
M --> N[Storyboard Engine]
M --> O[Image Prompt Engine]
M --> P[Video Prompt Engine]
M --> Q[TTS Engine]

```

```

M --> R[Caption Engine]
M --> S[Timeline + Render Engine]

```

5.3 Post-Production Architecture

```

flowchart TD
    A[Scene Master Validated] --> B[Media Asset Store]
    B --> C[Scene Video Assets]
    B --> D[Narration Audio]
    B --> E[Original Scene Audio]
    B --> F[Sound Effects]
    B --> G[Background Music]
    B --> H[Caption Text]

    C --> I[Timeline Assembly Engine]
    D --> J[Audio Mixing Engine]
    E --> J
    F --> J
    G --> J

    H --> K[Auto Caption Engine]
    I --> L[Render Composition Engine]
    J --> L
    K --> L

    L --> M[Render Validator]
    M --> N{Pass?}
    N -->|No| O[Repair Render / Missing Asset Fix]
    N -->|Yes| P[Final MP4 Export]

    P --> Q[CapCut Handoff Package]
    P --> R[Download / Share / Save Project]
    M --> S[Telemetry + Render Report]
    O --> S

```

5.4 GrokPI Provider Flow

```

sequenceDiagram
    participant UI as MASJAVAS UI
    participant API as Backend API
    participant Sec as Secret Manager
    participant GP as GrokPI API
    participant Pool as GrokPI Token Pool
    participant Up as Upstream Model
    participant Obs as Observability

    UI->>API: Generate request
    API->>Sec: Load GROKPI_API_KEY server-side

```



```
API-->>GP: GET /v1/models
GP-->>API: Allowed models
API-->>GP: POST /v1/chat/completions or /v1/video/generations or /v1/
audio/speech
GP-->>Pool: Select token by quota/status/cooldown
Pool-->>GP: Active token
GP-->>Up: Forward request
Up-->>GP: Response or job status
GP-->>API: Normalized response
API-->>Obs: Log provider event without secrets
API-->>UI: Result/status
```

6. Database Schema

6.1 Database Principles

- PostgreSQL untuk domain data.
- Object storage untuk upload, reference assets, video, audio, subtitles, final render.
- Provider secrets tidak boleh disimpan di domain logs.
- Semua critical action masuk audit_logs.
- Semua model/provider/render event masuk telemetry_events.
- Project history dan reusable reference menjadi retention foundation.

6.2 ERD

```
erDiagram
    users ||--o{ organization_members : joins
    organizations ||--o{ organization_members : has
    organizations ||--o{ projects : owns
    users ||--o{ projects : creates

    projects ||--o{ reference_assets : uploads
    projects ||--o{ ai_chat_messages : contains
    projects ||--o{ story_cores : has
    projects ||--o{ narrative_arcs : has
    projects ||--o{ narration_dna : has
    projects ||--o{ narration_masters : has
    narration_masters ||--o{ scene_narrations : contains
    narration_masters ||--o{ narration_validation_reports : validates

    projects ||--o{ universal_reference_sets : has
    universal_reference_sets ||--o{ reference_entities : contains
    projects ||--o{ scene_plans : has
    scene_plans ||--o{ scenes : contains
    scenes ||--o{ scene_beats : contains
    scenes ||--|| storyboard_prompts : has
    scenes ||--|| image_prompts : has
```

```

scenes ||--|| video_prompts : has
scenes ||--|| tts_prompts : has
scenes ||--|| scene_validation_reports : validates

projects ||--o{ provider_requests : executes
projects ||--o{ media_assets : stores
media_assets ||--o{ scene_video_assets : includes
media_assets ||--o{ audio_assets : includes
projects ||--o{ render_timelines : has
render_timelines ||--o{ timeline_video_clips : contains
render_timelines ||--o{ timeline_audio_tracks : contains
render_timelines ||--o{ caption_tracks : contains
projects ||--o{ render_jobs : has
render_jobs ||--o{ render_validation_reports : validates
projects ||--o{ export_packages : exports

projects ||--o{ feedback_events : receives
projects ||--o{ telemetry_events : logs
projects ||--o{ audit_logs : records
projects ||--o{ human_review_cases : escalates

users {
    uuid id PK
    string email
    string password_hash
    string name
    string status
    datetime created_at
    datetime updated_at
}

organizations {
    uuid id PK
    string name
    string plan
    string status
    datetime created_at
    datetime updated_at
}

organization_members {
    uuid id PK
    uuid organization_id FK
    uuid user_id FK
    string role
    datetime created_at
}

projects {
    uuid id PK
    uuid organization_id FK

```

```

    uuid owner_id FK
    string title
    string topic
    int total_scenes
    int duration_per_scene_sec
    int total_duration_sec
    string tone
    string platform
    string mode
    string status
    json metadata
    datetime created_at
    datetime updated_at
}

reference_assets {
    uuid id PK
    uuid project_id FK
    uuid uploaded_by FK
    string asset_type
    string label
    string file_url
    string storage_key
    json extracted_metadata
    string validation_status
    datetime created_at
}

story_cores {
    uuid id PK
    uuid project_id FK
    string story_core_hash
    json characters
    json opposing_forces
    json world
    json object_affordances
    json story_moments
    json risk_flags
    float completeness_score
    string status
    datetime created_at
}

narrative_arcs {
    uuid id PK
    uuid project_id FK
    uuid story_core_id FK
    json act_structure
    json scene_functions
    json hook_payoff_map
    float arc_quality_score

```

```

        string status
    }

    narration_dna {
        uuid id PK
        uuid project_id FK
        string narrator_voice
        string emotional_distance
        string moral_explicitness
        string metaphor_level
        json avoid_words
        json preferred_words
        json pacing_rules
        json tts_direction
        string status
    }

    narration_masters {
        uuid id PK
        uuid project_id FK
        uuid narrative_arc_id FK
        uuid narration_dna_id FK
        string version
        float human_like_score
        float repetition_score
        float speakability_score
        float timing_score
        float policy_score
        string status
        datetime locked_at
        datetime created_at
    }

    scene_narrations {
        uuid id PK
        uuid narration_master_id FK
        int scene_number
        text narration
        text dialogue
        string emotional_function
        string subtext
        float estimated_read_time_sec
        json voice_notes
        string status
    }

    narration_validation_reports {
        uuid id PK
        uuid narration_master_id FK
        string validation_type
        float score
    }

```

```

        json issues
        json repair_suggestions
        string status
        datetime created_at
    }

    universal_reference_sets {
        uuid id PK
        uuid project_id FK
        string story_core_hash
        json validation
        string status
        datetime created_at
    }

    reference_entities {
        uuid id PK
        uuid reference_set_id FK
        uuid reference_asset_id FK
        string reference_id
        string role
        string canonical_name
        string source_story_core_entity_id
        json usage_rules
        text image_prompt
        text negative_prompt
        json validation_rules
        string status
    }

    scene_plans {
        uuid id PK
        uuid project_id FK
        uuid narration_master_id FK
        int total_scenes
        int duration_per_scene_sec
        string status
    }

    scenes {
        uuid id PK
        uuid scene_plan_id FK
        uuid project_id FK
        int scene_number
        string scene_id
        string title
        int duration_sec
        string story_action
        string status
    }
}

```

```

scene_beats {
    uuid id PK
    uuid scene_id FK
    string time_range
    text beat_description
    string beat_type
}

storyboard_prompts {
    uuid id PK
    uuid scene_id FK
    text prompt
    json references
    string status
}

image_prompts {
    uuid id PK
    uuid scene_id FK
    text prompt
    text negative_prompt
    json reference_ids
    float reference_compliance_score
}

video_prompts {
    uuid id PK
    uuid scene_id FK
    text prompt
    json reference_ids
    json beat_map
    json camera
    json motion
    json sound
    float validation_score
}

tts_prompts {
    uuid id PK
    uuid scene_id FK
    text narration_text
    text dialogue_text
    string voice
    json pacing
    json emotion
    float audio_validation_score
}

media_assets {
    uuid id PK
    uuid project_id FK

```

```

    string asset_type
    string storage_url
    string source
    json metadata
    string status
    datetime created_at
}

scene_video_assets {
    uuid id PK
    uuid media_asset_id FK
    uuid scene_id FK
    int scene_number
    float duration_sec
    string validation_status
}

audio_assets {
    uuid id PK
    uuid media_asset_id FK
    string audio_type
    float duration_sec
    float loudness_lufs
    string validation_status
}

render_timelines {
    uuid id PK
    uuid project_id FK
    int total_scenes
    int duration_per_scene_sec
    int total_duration_sec
    json render_settings
    string status
    datetime created_at
}

timeline_video_clips {
    uuid id PK
    uuid render_timeline_id FK
    int scene_number
    uuid asset_id FK
    float start_sec
    float end_sec
    float duration_sec
    string transition_in
    string transition_out
}

timeline_audio_tracks {
    uuid id PK

```

```

        uuid render_timeline_id FK
        string track_id
        string track_type
        json clips
        float volume_db
        bool ducking
    }

    caption_tracks {
        uuid id PK
        uuid render_timeline_id FK
        string source
        string format
        json captions
        string validation_status
    }

    render_jobs {
        uuid id PK
        uuid project_id FK
        uuid render_timeline_id FK
        string status
        string output_url
        string error_code
        json settings
        datetime started_at
        datetime completed_at
    }

    provider_requests {
        uuid id PK
        uuid project_id FK
        uuid scene_id FK
        string provider
        string endpoint
        string model
        string request_id
        string job_id
        string status
        string error_code
        int duration_ms
        int tokens_input
        int tokens_output
        float cost_estimate
        datetime created_at
    }

    export_packages {
        uuid id PK
        uuid project_id FK
        uuid render_job_id FK
    }

```



```

    string package_type
    string storage_url
    string status
    datetime created_at
}

```

7. Design & Technical Constraints

7.1 UX Constraints

Constraint	Requirement
First-run UX	User harus bisa membuat project pertama dengan input sedikit.
Scene-count UX	Total scene harus jelas, total durasi otomatis terlihat.
Narration-first UX	UI harus menampilkan bahwa narasi adalah gate utama.
Narration review UX	User bisa review, approve, repair, atau ubah style narator sebelum visual dibuat.
Upload UX	Reference upload harus bisa diberi label: character, location, style, source, moodboard.
Chat UX	Assistant harus memberi next action, bukan hanya jawaban teks.
Pipeline UX	Phase, blockers, confidence, dan validation harus terlihat.
Timeline UX	User melihat urutan scene dan status asset.
Audio UX	User melihat track narasi, scene audio, SFX, music, dialogue.
Caption UX	User bisa pilih no caption, SRT-only, burn-in, atau both.
Render UX	User bisa preview scene, preview segment, draft render, final render.
Export UX	Export disabled jika validation fail.

7.2 Technical Constraints

Layer	Recommendation
Frontend	Next.js / React / TypeScript.
Backend	Node/NestJS or Python/FastAPI.
Orchestration	State machine + queue-based jobs.
Queue	BullMQ/Redis or Celery.
Database	PostgreSQL.
Object Storage	S3-compatible.

Layer	Recommendation
Provider	GrokPI OpenAI-compatible gateway.
Rendering	FFmpeg-based render service or dedicated media worker.
Captions	SRT/VTT/ASS generator.
Audio Mixing	FFmpeg/audio processing worker.
Observability	Structured logs, traces, metrics, alerts.
Deployment MVP	Docker Compose.
Scale	Kubernetes/container workers.
Secrets	Secret manager/env server-side.

7.3 Security Constraints

- API key GrokPI tidak boleh berada di frontend.
- Authorization header tidak boleh masuk logs.
- Uploaded files harus divalidasi tipe, ukuran, permission, dan malware risk.
- User hanya dapat mengakses project dan reference miliknya.
- PII/secrets dalam dokumen upload harus dimasking bila masuk model call.
- Export package harus melewati secret leakage check.
- Render logs tidak boleh menyimpan raw secret.

7.4 GrokPI Environment Contract

```
AI_PROVIDER=grokpi
GROKPI_BASE_URL=https://www.grokpi.masjavas.my.id
GROKPI_API_KEY=<set-in-secret-manager>
GROKPI_DEFAULT_TEXT_MODEL=grok-4.1-expert
GROKPI_FAST_TEXT_MODEL=grok-4.1-fast
GROKPI_IMAGE_MODEL=grok-image-1.0
GROKPI_IMAGE_FAST_MODEL=grok-image-1.0-fast
GROKPI_IMAGE_EDIT_MODEL=grok-image-1.0-edit
GROKPI_VIDEO_MODEL=grok-image-1.0-video
GROKPI_TTS_MODEL=grok-tts
```

8. AI Reliability Layer

8.1 Reliability Principle

MASJAVAS harus dirancang sebagai sistem AI yang probabilistic, degradable, non-deterministic, dan failure-prone. Karena itu, setiap output AI wajib memiliki:

- Guardrails

- Confidence scoring
- Fallback hierarchy
- Human escalation
- Observability
- Evaluation
- Rollback condition
- Release threshold

8.2 Guardrails

Layer	Guardrail	Trigger	Action
Input	Required field validation	Topic/totalScenes kosong	Block request
Input	Scene count validation	totalScenes invalid	Ask correction
Input	File validation	Unsupported/unsafe upload	Reject file
Input	Permission filtering	User akses file bukan miliknya	Deny
Input	Prompt injection detection	User mencoba override system rules	Reject/sanitize
Input	PII/secret detection	Sensitive data in upload	Mask/ask consent
Story	StoryCore completeness	Score rendah	Ask clarification
Narration	Anti-AI tone	Narasi generik/repetitif	Rewrite
Narration	Visual-description overload	Narasi hanya menjelaskan visual	Rewrite
Narration	Timing violation	Narasi terlalu panjang	Shorten/rewrite
Narration	Policy/genre violation	Unsafe or genre leak	Block/repair
Reference	Role misuse	Character ref dipakai sebagai action scene	Repair prompt
Reference	Identity drift	Face/wardrobe mismatch	Repair/ref upload
Scene	Beat map missing	No 0–10s map	Block scene
Video	Subtitle/visible text risk	Prompt contains visible text	Remove/block
Provider	Model unavailable	model_not_found	Sync/fallback
Provider	Rate limit	429/daily limit	Backoff/queue/admin alert
Timeline	Missing scene asset	Scene video not found	Block render
Audio	Audio starts before 2s	Timing violation	Rebuild audio map
Caption	Caption out of sync	Caption drift	Rebuild caption

Layer	Guardrail	Trigger	Action
Render	Final duration mismatch	Duration fail	Block export
Release	Validation fail	Any critical fail	No export

8.3 Confidence Scoring

Signal	Weight
StoryCore completeness	10%
Narrative Arc quality	10%
Narration human-like quality	20%
Narration repetition score	10%
Narration timing/speakability	10%
Reference consistency	10%
Scene plan validity	10%
Provider health	5%
Timeline/render validation	10%
Safety/security	5%

8.4 Fallback Hierarchy

Primary GrokPI model/task provider

- Fallback model within same capability
- Prompt/narration repair mode
- Rule-based validator
- Human review
- Graceful failure with no export

8.5 Provider Error Handling

Error Code	Handling
invalid_json	Block request; log payload validation failure.
missing_model	Repair payload; do not call provider.
invalid_messages	Repair message schema.
invalid_image_config	Repair image_config.
invalid_video_config	Repair video_config.

Error Code	Handling
missing_prompt	Block media request; rebuild prompt.
missing_image	Ask for image_url/reference upload.
invalid_api_key	Stop execution; alert admin.
model_not_allowed	Use allowed fallback or request whitelist update.
media_generation_disabled	Block media workflow; notify admin.
model_not_found	Sync /v1/models; fallback model.
rate_limit_exceeded	Retry with backoff; queue if safe.
daily_limit_exceeded	Stop; quota notice and admin escalation.

9. Evaluation, Observability & Governance

9.1 Evaluation Categories

Category	Purpose
Activation success	User berhasil membuat project pertama.
StoryCore quality	Ide berhasil diekstrak menjadi story elements.
Narrative Arc quality	Cerita terbagi proporsional sesuai scene count.
Narration human-like quality	Narasi tidak terasa dibuat AI.
Narration repetition	Tidak repetitif antar scene.
Narration speakability	Enak dibacakan manusia/TTS.
Narration timing	Cocok dengan scene 10 detik.
Reference consistency	Karakter/lokasi/style konsisten.
Prompt quality	Image/video separation dan reference compliance.
Video quality	10-second event, no visible text, continuity.
Audio quality	Narration/dialogue timing, loudness, ducking.
Caption quality	Subtitle sync dan format valid.
Timeline quality	Scene order dan completeness.
Render quality	Final MP4 valid dan durasi benar.
Provider reliability	GrokPI success/error/latency/job completion.
Safety/security	Secret leakage, PII, policy violation.
Regression	Update prompt/model tidak merusak output lama.

9.2 Golden Dataset

Field	Description
case_id	Unique test case ID
user_persona	Creator/documentary/short-form/agency
input_topic	Topic mentah user
total_scenes	Jumlah scene target
uploaded_reference	Character/source/style files
expected_storycore	Expected extracted story structure
expected_narration_style	Human-like narration target
ideal_output	Reference package/render behavior
risk_level	Low/Medium/High/Critical
evaluation_rubric	Rubric penilaian
must_not_do	Hal yang dilarang

Dataset split:

- 40% common creator workflow.
- 20% narration quality edge cases.
- 15% reference consistency cases.
- 10% post-production/render cases.
- 10% adversarial/prompt injection.
- 5% provider failure simulation.

9.3 Metrics

Metric	Target	Rollback
Project activation rate	$\geq 75\%$	$< 60\%$
Narration Master Lock success	$\geq 70\%$	$< 55\%$
Human-like narration score	≥ 95	< 90
Narration repetition score	≥ 95	< 90
Speakability score	≥ 95	< 90
Narration timing pass	$\geq 95\%$	$< 90\%$
Scene count accuracy	100%	$< 100\%$
Scene duration accuracy	100%	$< 100\%$
Reference usage success	$\geq 90\%$	$< 80\%$
Reference consistency	$\geq 95\%$	$< 90\%$

Metric	Target	Rollback
Prompt separation	100%	< 100%
Raw video visible text violation	0	> 0
Audio sync pass	$\geq 95\%$	< 90%
Caption sync pass	$\geq 95\%$	< 90%
Timeline assembly success	$\geq 95\%$	< 90%
Render success rate	$\geq 95\%$	< 90%
Final duration accuracy	100%	< 100%
Provider error rate	$\leq 3\%$	> 5%
Safety violation rate	$\leq 0.5\%$	> 1%
Regression pass rate	$\geq 95\%$	< 90%

9.4 Required Observability Events

```

request_received
user_authenticated
project_created
scene_count_validated
reference_upload_started
reference_upload_completed
reference_upload_failed
chat_message_received
storycore_extraction_started
storycore_extraction_completed
narrative_arc_created
narration_dna_created
narration_generation_started
narration_generation_completed
anti_ai_tone_check_completed
repetition_check_completed
narration_timing_check_completed
narration_policy_check_completed
narration_master_locked
narration_repair_started
narration_repair_completed
universal_reference_set_created
reference_validation_passed
reference_validation_failed
scene_plan_created
beat_map_created
storyboard_prompt_generated
image_prompt_generated
video_prompt_generated

```

```

tts_prompt_generated
scene_master_created
approval_requested
approval_granted
provider_model_sync_started
provider_model_sync_completed
provider_request_started
provider_request_completed
provider_request_failed
video_job_submitted
video_job_status_updated
video_job_completed
video_job_failed
scene_video_asset_validated
timeline_assembly_started
timeline_assembly_completed
audio_mix_started
audio_mix_completed
caption_generation_started
caption_generation_completed
render_started
render_completed
render_failed
render_validation_passed
render_validation_failed
export_blocked
export_completed
fallback_invoked
human_escalation_created
incident_detected
user_feedback_received
rollback_triggered

```

9.5 Dashboards

Dashboard	Metrics
Product Health	Project created, activation rate, first render success, repeat usage.
Narration Quality	Human-like score, repetition score, repair rate, timing fail, user rating.
Reference Health	Upload success, reference reuse, identity drift, role misuse.
Prompt/Scene Quality	Prompt separation, scene validation, beat map pass.
Provider Health	GrokPI sync, request status, video job status, rate limit.
Post-Production Health	Timeline success, audio sync, caption sync, render success.
Cost/Quota	Tokens, video jobs, TTS jobs, render time, storage.
Incidents	Export blocked, provider failure, render failure, secret alerts.

9.6 Governance Roles

Role	Responsibility
Product Owner	User value, activation, retention, roadmap.
Narration Owner	Narration quality, DNA, style, script doctor, anti-AI tone.
System Owner	Architecture, reliability, uptime, export gate.
Model/Prompt Owner	Prompt templates, model routing, evals, regression.
Provider Owner	GrokPI keys, token pool, quota, uptime, model sync.
Media Owner	Scene video, image, TTS, audio, asset validation.
Render Owner	Timeline, mixing, caption, render, post-production validation.
Data Owner	Uploaded references, retention, access control.
Security Owner	Secrets, PII, permission, secure logs.
Risk Owner	Risk classification, failure simulation, release gating.
Human Review Owner	Escalation queue and approval.
Release Approver	Go/no-go release decision.

9.7 Governance Policies

- Every AI phase must have an owner.
- Narration is the first creative release gate.
- No visual/media artifact may be generated before Narration Master Lock.
- Every prompt/model/provider/render config must be versioned.
- Every export must have validation proof.
- Every secret must be masked and server-side only.
- Every provider failure must be logged with error_code but without sensitive data.
- Every render must pass timeline/audio/caption/duration validation.
- Every release must pass eval and rollback thresholds.

10. Rollout, Risks & Release Thresholds

10.1 Rollout Strategy

Phase	Scope	Goal	Success Criteria	Exit Criteria
Prototype	Internal builder	Validate project + scene count + narration flow	Project + Narration Master works	Stakeholder approval
Internal Alpha	Internal creators/testers	Validate narration-first pipeline and reference system	No critical blockers; narration lock works	Eval baseline passed

Phase	Scope	Goal	Success Criteria	Exit Criteria
Media Alpha	Internal video tests	Validate video asset + timeline + audio + caption	Draft render works	Render validation stable
Private Beta	20–50 selected creators	Validate usefulness and professional quality	Task success $\geq 80\%$; narration rating $\geq 4/5$	Feedback triaged
Public MVP	Wider creator users	Validate repeat usage and render	Task success $\geq 85\%$; render success $\geq 95\%$	Monitoring stable
Pro Release	Paid power users	Validate routine production use	Repeat usage $\geq 35\%$; provider errors $\leq 3\%$	Cost/quota controlled
Scale	Team/agency users	Add team workflow and advanced post-production	SLA stable; governance mature	Enterprise roadmap defined

10.2 Implementation Roadmap

Sprint	Focus	Deliverables
1	Foundation	Auth, workspace, project creation, scene-count-first config.
2	Upload Reference	File validation, storage, metadata, reference labeling.
3	AI Assistant	Project-aware chat, brief builder, next action.
4	StoryCore + Arc	StoryCore extractor, narrative arc planner.
5	Narration DNA	Narration persona, style rules, pacing rules.
6	Narration Master	Scene narration generator, script doctor, lock flow.
7	Narration Validators	Anti-AI tone, repetition, timing, speakability, policy.
8	Universal Reference Set	Planner, prompt generator, role validator.
9	Scene Plan + Beat Map	10s beat map, scene plan, scene validators.
10	Prompt Engines	Storyboard, image prompt, video prompt, TTS prompt.
11	GrokPI Adapter	Model sync, text/image/video/TTS, error mapping.
12	Scene Master + Approval	Scene sheet, user approval, repair mode.
13	Media Assets	Scene video storage, asset manifest, duration validator.
14	Caption + Audio	Caption generator, audio track builder, audio mixing.
15	Timeline Engine	Scene ordering, timeline JSON, validation.
16	Render Engine	Draft render, final render, render validator.

Sprint	Focus	Deliverables
17	Export Package	Final MP4, subtitles, timeline, CapCut handoff.
18	Observability + Beta	Dashboards, alerts, evals, feedback loop.

10.3 Risk Register

Risk	Severity	Likelihood	Impact	Mitigation	Owner
Narasi terasa seperti AI	Critical	Medium	Film kehilangan nyawa	Narration DNA + Script Doctor + Anti-AI Detector	Narration Owner
Narasi repetitif	High	Medium	Penonton bosan	Repetition Detector + arc function map	Narration Owner
Narasi terlalu visual/deskriptif	High	Medium	Tidak cinematic	Visual-description guardrail	Narration Owner
Narasi terlalu panjang untuk 10 detik	High	Medium	Audio tidak sinkron	Timing Validator + rewrite	Audio Owner
User tetap merasa workflow kompleks	High	Medium	Activation turun	Guided onboarding + Assistant	Product Owner
Uploaded reference buruk	Medium	High	Output drift	Reference quality check	Data Owner
Character/ reference drift	High	Medium	Output tidak profesional	Reference Lock + validator	Model Owner
Prompt image/ video tercampur	High	Medium	Media output buruk	Prompt separation validator	Model Owner
Scene video missing	High	Medium	Timeline gagal	Asset validator + repair	Media Owner
Audio mix buruk	High	Medium	Output tidak profesional	Loudness + ducking validator	Render Owner
Caption tidak sinkron	Medium	Medium	Viewer experience buruk	Caption sync validator	Render Owner
Final duration salah	High	Medium	Render tidak valid	Render duration validator	Render Owner
GrokPI provider down/error	High	Medium	Generation gagal	Fallback, queue, admin alert	Provider Owner

Risk	Severity	Likelihood	Impact	Mitigation	Owner
Secret leakage	Critical	Low	Security incident	Secret manager + masking + export scan	Security Owner
Export output belum valid	Critical	Low	User trust loss	Export gate	System Owner
Hallucination documentary	Critical	Medium	Misinformation	Source confidence + human review	Risk Owner
Cost/quota membengkak	Medium	Medium	Margin buruk	Usage metering + quota dashboard	Provider Owner
User tidak balik lagi	High	Medium	Retention buruk	Project history, reusable references, render history	Product Owner

10.4 Release Thresholds

Release hanya boleh dilakukan jika:

- Critical bugs = 0.
- High-risk unresolved issues = 0.
- User bisa membuat project pertama end-to-end.
- Scene count accuracy = 100%.
- Scene duration accuracy = 100%.
- Narration Master Lock success $\geq 70\%$.
- Human-like narration score ≥ 95 .
- Repetition score ≥ 95 .
- Speakability score ≥ 95 .
- Narration timing pass $\geq 95\%$.
- Policy/safety pass = 100% untuk high-risk cases.
- Upload reference success $\geq 90\%$.
- Reference consistency $\geq 95\%$.
- Image/video separation = 100%.
- Raw video visible text violation = 0.
- Timeline assembly success $\geq 95\%$.
- Audio sync pass $\geq 95\%$.
- Caption sync pass $\geq 95\%$.
- Render success rate $\geq 95\%$.
- Final duration accuracy = 100%.
- Provider error rate $\leq 3\%$.
- Fallback success rate $\geq 95\%$.
- Safety violation rate $\leq 0.5\%$.
- Regression pass rate $\geq 95\%$.
- No API key/token exposure in frontend/log/export.
- Human escalation path ready.
- Incident runbook ready.

10.5 Rollback Conditions

Rollback wajib jika:

- Narration Master Lock menghasilkan output yang human-like score < 90.
- Repetition score < 90.
- Scene count accuracy < 100%.
- Scene duration accuracy < 100%.
- Raw video menghasilkan visible text/subtitle tidak disengaja.
- Export gate gagal memblokir invalid output.
- API key/token bocor.
- Reference consistency < 90%.
- Image/video separation < 100%.
- Audio sync pass < 90%.
- Caption sync pass < 90%.
- Render success < 90%.
- Final duration accuracy < 100%.
- Provider error rate > 5%.
- Fallback success rate < 90%.
- Safety violation rate > 1%.
- Regression pass rate < 90%.
- User activation turun < 60% setelah release.
- Human escalation backlog melebihi SLA.

10.6 MVP Acceptance Criteria

- [] User dapat login dan membuat workspace.
- [] User dapat membuat project dari TOPIC, TOTAL_SCENES, TONE, PLATFORM, MODE.
- [] Sistem menghitung totalDuration = totalScenes × 10 detik.
- [] User dapat upload file/gambar/dokumen sebagai reference.
- [] AI Assistant dapat membantu brief, repair, dan next action.
- [] StoryCore Extractor menghasilkan StoryCore.
- [] Narrative Arc Planner membagi cerita sesuai jumlah scene.
- [] Narration DNA dibuat dan dikunci.
- [] Narration Master dibuat, divalidasi, dan dilock sebelum artefak lain.
- [] Anti-AI Tone Detector berjalan.
- [] Repetition Detector berjalan.
- [] Narration Timing Validator berjalan.
- [] Universal Reference Set dibuat dari StoryCore.
- [] Reference roles divalidasi.
- [] Scene Plan dan Beat Map 10 detik dibuat.
- [] Storyboard, image prompt, video prompt, dan TTS prompt dibuat setelah Narration Master pass.
- [] GrokPI model sync berjalan via `/v1/models`.
- [] Text/orchestration berjalan via `/v1/chat/completions`.
- [] Video generation berjalan via `/v1/video/generations` dengan polling.
- [] TTS berjalan via `/v1/audio/speech`.
- [] Scene video asset tersimpan dan divalidasi.
- [] Timeline Assembly Engine mengurutkan scene.
- [] Audio Mixing Engine menggabungkan narration, original audio, SFX, music, dialogue.

- [] Auto Caption Engine membuat SRT/VTT/ASS/JSON.
- [] Render Composition Engine menghasilkan final MP4.
- [] Render Validator memastikan final duration benar.
- [] Export diblokir jika validation fail.
- [] CapCut Handoff Package tersedia.
- [] Project history, reusable reference, dan render history tersedia.
- [] Observability events tercatat untuk critical path.
- [] Secrets tidak terekspos di frontend, logs, export, atau PDF.

Appendix A — Internal Narration Prompt Template

You are a professional Indonesian cinematic narration writer.

Goal:

Write scene narration that feels human, cinematic, restrained, emotionally specific, and speakable.

Rules:

- Do not describe obvious visuals.
- Transfer emotion, tension, meaning, or subtext.
- Avoid generic AI phrasing.
- Avoid moral lecture.
- Avoid repeated sentence patterns.
- Avoid overdramatic language.
- Use natural Indonesian.
- Fit scene timing: narration starts at 2.0s.
- Must not include subtitles, camera instruction, or visual prompt.
- Must match Narration DNA.
- Must preserve story continuity.
- Must pass policy and genre guardrails.

Input:

StoryCore:

...

Narrative Arc:

...

Narration DNA:

...

Scene Function:

...

Previous Scene Narration:

...

Next Scene Goal:

...

Output JSON:

```
{
  "sceneNumber": 1,
  "narration": "...",
  "dialogue": "None or short line",
  "emotionalFunction": "...",
  "subtext": "...",
  "forbiddenRepeatsChecked": true,
  "speakabilityNotes": "...",
  "estimatedReadTimeSec": 3.2
}
```

Appendix B — Timeline Manifest Example

```
{
  "projectId": "uuid",
  "totalScenes": 30,
  "durationPerSceneSec": 10,
  "totalDurationSec": 300,
  "timeline": {
    "videoTracks": [
      {
        "trackId": "V1",
        "type": "main_scene_video",
        "clips": []
      },
      {
        "trackId": "V2",
        "type": "overlay_caption_or_graphic",
        "clips": []
      }
    ],
    "audioTracks": [
      {
        "trackId": "A1",
        "type": "narration",
        "clips": []
      },
      {
        "trackId": "A2",
        "type": "original_scene_audio",
        "clips": []
      },
      {
        "trackId": "A3",
        "type": "sound_effects",
        "clips": []
      }
    ]
  }
}
```

```

    },
    {
      "trackId": "A4",
      "type": "background_music",
      "clips": []
    },
    {
      "trackId": "A5",
      "type": "dialogue",
      "clips": []
    }
  ],
  "captionTracks": [
    {
      "trackId": "C1",
      "type": "subtitle",
      "format": "srt_vtt_ass"
    }
  ]
}
}

```

Appendix C — Implementation Module Map

```

src/lib/project/
- project-builder.ts
- scene-count-adapter.ts

src/lib/storycore/
- storycore-extractor.ts
- narrative-arc-planner.ts

src/lib/narration/
- narration-dna-builder.ts
- narration-master-generator.ts
- scene-narration-generator.ts
- script-doctor-rewriter.ts
- anti-ai-tone-detector.ts
- repetition-detector.ts
- emotional-continuity-validator.ts
- narration-policy-guardrail.ts
- narration-timing-validator.ts
- narration-quality-scorer.ts

src/lib/references/
- universal-reference-planner.ts
- universal-reference-prompt-generator.ts
- universal-reference-validator.ts

```



```
src/lib/scene/
- scene-planner.ts
- beat-map-generator.ts
- scene-validator.ts
- scene-master-builder.ts

src/lib/prompts/
- storyboard-prompt-generator.ts
- image-prompt-generator.ts
- video-prompt-generator.ts
- tts-prompt-generator.ts

src/lib/provider/
- grokpi-client.ts
- model-sync.ts
- provider-router.ts
- provider-error-mapper.ts
- async-video-poller.ts

src/lib/media/
- media-asset-manager.ts
- scene-video-validator.ts
- asset-manifest-generator.ts

src/lib/timeline/
- timeline-assembler.ts
- scene-order-validator.ts
- timeline-manifest-exporter.ts

src/lib/audio/
- narration-track-builder.ts
- audio-mixer.ts
- ducking-engine.ts
- loudness-validator.ts

src/lib/caption/
- caption-generator.ts
- srt-exporter.ts
- vtt-exporter.ts
- ass-exporter.ts
- caption-sync-validator.ts

src/lib/render/
- render-composition-engine.ts
- render-preview-engine.ts
- final-render-engine.ts
- render-validator.ts

src/lib/export/
```

- `capcut-handoff-exporter.ts`
- `final-package-exporter.ts`